

PROJECTED MULTILINEAR TREES: EXACT ERROR DECOMPOSITION, UNIVERSAL BOUNDS, AND INDEPENDENT-LAW SHARPNESS

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ABSTRACT. We study finite typed trees whose internal vertices apply multilinear laws and whose recursively evaluated outputs are orthogonally projected onto reduced subspaces. An exact child-error expansion separates local closure leakage from propagated and higher-order branch interactions. This yields the universal ambient and projected-root coefficients k and $k - 1$. We close the independent-law $k = 2$ sharpness question, give a necessary-and-sufficient equality characterization valid for independent or repeated laws, and solve an explicitly defined contractive gated repeated-law subclass. For $k = 3$ we provide certified lower witnesses, an unconditional tightened upper envelope, a corrected chain-only non-attainment argument, and the exact independent-law fixed- η constant W_3 for every finite arity profile. Finally, every fixed finite ordered rooted topology with arities at least two is shown to be asymptotically sharp with coefficient $k - 1$ as $\eta \downarrow 0$. Same-law/gated, $k \geq 4$ fixed- η , and growing-tree variants remain open.

1. SCOPE AND EPISTEMIC STATUS

This paper is the mathematical core of a larger repository. Executable behavior is governed by the source and tests; mathematical status is governed by the theorem registries and proof dossiers. Numerical residuals and finite checks are reported as evidence, not promoted to universal theorems. The finite computational core is frozen at commit `1f4984ec8e741049789e0035c7a3ba84c86d3f29`; the v5 theorem package is recorded at the subsequent scientific commits listed in the reproducibility companion.

Throughout, “independent laws” means that the bounded multilinear law at each internal vertex is freely selectable, subject only to the declared operator-norm and local closure-defect budgets. “Same law” means that one bilinear map is reused at every occurrence of the declared topology; it does not by itself impose rank-one projectors or common leaves. “Fixed tree” and “finite arity” always mean a single finite ordered rooted topology, while the limits in this paper are taken only after that topology has been fixed. These distinctions are part of every extremal constant’s definition.

2. TYPED PROJECTED TREES

Let $\mathcal{T}$ be a finite rooted tree. For every type τ , let V_τ and W_τ be finite-dimensional real or complex Hilbert spaces and let $Q_\tau : W_\tau \rightarrow V_\tau$ be an isometry. Put

$$P_\tau = Q_\tau Q_\tau^*, \quad P_\tau^2 = P_\tau = P_\tau^*.$$

An internal vertex v of arity a_v carries a bounded typed law

$$\mu_v : \prod_{j=1}^{a_v} V_{\tau(v,j)} \longrightarrow V_{\tau(v)}.$$

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Research manuscript. The theorem statuses in this paper are scoped to the stated hypotheses. Independent human review and theorem-level novelty review remain pending.

Write $M_v = \|\mu_v\|_{\text{op}}$ and define the local closure map

$$r_v = (I - P_v)\mu_v(P_{c_1}\cdot, \dots, P_{c_{a_v}}\cdot), \quad \rho_v = \|r_v\|_{\text{op}}.$$

For reduced leaves $z_\ell \in W_{\tau(\ell)}$, define the ambient and recursively projected evaluations by

$$F_\ell = R_\ell = Q_{\tau(\ell)}z_\ell, \quad F_v = \mu_v(F_{c_1}, \dots, F_{c_{a_v}}), \quad R_v = P_v\mu_v(R_{c_1}, \dots, R_{c_{a_v}}).$$

At the root, define

$$E^A = \|F - R\|, \quad E^P = \|PF - R\|, \quad E^N = \|(I - P)F\|, \quad E^{\text{red}} = \|Q^*F - Q^*R\|.$$

Proposition 2.1 (Exact root geometry). *For every valid typed tree,*

$$(E^A)^2 = (E^P)^2 + (E^N)^2, \quad E^{\text{red}} = E^P.$$

The first identity is the orthogonal decomposition $F - R = (PF - R) + (I - P)F$; the second follows from the isometry of Q on the projected range.

3. EXACT LOCAL ERROR ALGEBRA

Let $D_v = F_v - R_v$, and for a nonempty subset S of child slots let $y_i^S = D_i$ when $i \in S$ and $y_i^S = R_i$ otherwise. Multilinearity gives the exact identity

$$(1) \quad D_v = r_v(R_1, \dots, R_{a_v}) + \sum_{\emptyset \neq S \subseteq [a_v]} \mu_v(y_1^S, \dots, y_{a_v}^S).$$

Applying P_v annihilates the new local normal residual because $P_v(I - P_v) = 0$. Equation (1) is an identity, not a bound; the inequalities begin only after taking norms and applying operator estimates.

4. UNIVERSAL COEFFICIENTS

Let k be the number of internal vertices and $L = \prod_\ell \|z_\ell\|$. Under $M_v \leq M$ and $\rho_v \leq \rho$, induction on the tree gives

Theorem 4.1 (Universal ambient and projected bounds).

$$E^A \leq k\rho M^{k-1}L, \quad E^N \leq k\rho M^{k-1}L, \quad E^P = E^{\text{red}} \leq (k-1)\rho M^{k-1}L.$$

The root source contributes to ambient and normal error but not to the projected-root error. This is the exact origin of the coefficient change $k \rightarrow k-1$. The theorem is conditional only on the finite-dimensional typed Hilbert-space, orthogonal-projector, multilinearity, and uniform norm/defect hypotheses stated above.

5. FIXED-ETA SHARPNESS FOR TWO-NODE CHAINS

For a fixed binary chain define the normalized projected constant

$$C_T^P(\eta) = \sup \frac{E^P}{\rho ML}, \quad \eta = \rho/M.$$

The supremum is over the declared admissible class. The universal theorem implies $C_T^P(\eta) \leq 1$ for $k = 2$.

Theorem 5.1 (Sharpness for independent binary laws). *Consider real two-dimensional spaces, the rank-one projector $P = \text{diag}(1, 0)$, unit leaves e_0, e_0, e_0 , and independently chosen bilinear laws. For every $0 < \eta \leq 1$, there are laws with $\|\mu_1\|_{\text{op}} = \|\mu_2\|_{\text{op}} = M$, closure defects at most $\rho = \eta M$, and*

$$E^P = \rho M.$$

Consequently,

$$\boxed{C_{2,\text{ind}}^P(\eta) = 1}, \quad 0 < \eta \leq 1.$$

Proof. Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. Define

$$\mu_1(x, y) = Mx_1y_1e_0 + \rho x_0y_0e_1, \quad \mu_2(x, y) = Mx_1y_0e_0.$$

For unit x, y , the two output components of μ_1 have squared norm at most $M^2(x_1^2y_1^2 + \eta^2x_0^2y_0^2) \leq M^2$; equality is attained on the normal-normal input. The second law has norm M and the same estimate is immediate. On projected inputs, the first closure map has norm ρ and the second has norm zero. The ambient first-node value on (e_0, e_0) is ρe_1 , while its projected value is zero. The second law maps this normal error with gain M into $\text{ran } P$. Hence $E^P = M\rho$. The upper bound in Theorem 4.1 supplies the matching inequality. $\square$

Remark 5.2 (Parameter sharing changes the extremal problem). The earlier gated-planar/repeated-law construction gives $E^P = \eta^2 M^2$, hence only the lower bound $C_{2,\text{rep}}^P(\eta) \geq \eta$ after normalization. The universal upper remains $C_{2,\text{rep}}^P(\eta) \leq 1$. Thus independent-law and repeated-law sharpness are distinct extremal problems.

There is, however, a canonical broader gated subclass for which the repeated law can be optimized exactly. Let $\mu_A(x, y) = Ax\langle e_0, y \rangle$, with A supported on the active two-plane, $\|A\| \leq M$, and $\|(I - P)AP\| \leq \rho$. The exact two-node identity is

$$PA^2e_0 - PAP Ae_0 = PA(I - P)Ae_0.$$

Consequently $E^P \leq M\rho$, and the contractive off-diagonal matrix $A = \begin{pmatrix} 0 & M \\ \rho & 0 \end{pmatrix}$ attains equality. Hence this explicitly defined contractive gated-planar repeated subclass has normalized constant one; the value η^2 is a consequence of the additional orthogonality constraint in the canonical rotation family. The proof and evaluator are recorded in `research/math_closure/k2/m17_broader_gated_contraction_exact.tex`.

6. THE $k = 2$ EQUALITY CHARACTERIZATION

The construction above proves sharpness, but the equality mechanism can be classified without choosing a particular coordinate model. Consider a binary chain with inner input (a, b) , outer input (D, d) , and $D = (I - P_{\text{in}})\mu_{\text{in}}(a, b)$. Equality in the projected-root theorem requires equality in every link of the proof chain.

Theorem 6.1 (Necessary and sufficient conditions for $k = 2$ saturation). *The identity $E^P = \rho ML_T$ holds if and only if all three conditions hold:*

- (EQ1) μ_{out} attains its norm bound M on (D, d) ;
- (EQ2) the inner closure map attains its bound ρ on (a, b) ;
- (EQ3) $\mu_{\text{out}}(D, d)$ lies in the range of the root projector.

This statement allows independent or repeated bilinear laws, arbitrary finite dimension, and arbitrary orthogonal-projector rank.

The proof is an equality-of-endpoints argument applied to the elementary chain of inequalities: if the endpoint equals the product of the declared upper bounds, every intermediate inequality is an equality. Conversely, EQ1–EQ3 make every link tight. The proof dossier and implementation cross-check are recorded in `research/math_closure/k2/saturation_iff_theorem.tex` and `src/seion_core/research_v5/k2_characterization.py`. This theorem does not assert sharpness for every restricted subclass: the specific gated-planar family has the separate exact value $E^P = \eta^2$.

7. INDEPENDENT-LAW $k=3$ LOWER WITNESSES

The `v5` construction package supplies separate chain and branching witnesses. With $q = \min(\rho, M/\sqrt{2})$, both yield

$$E^P = 2Mq\sqrt{M^2 - q^2}.$$

Therefore, relative to the $k = 3$ normalization ρM^2 ,

$$C_{3,\text{ind}}^P(\eta) \geq \begin{cases} 2\sqrt{1-\eta^2}, & 0 < \eta \leq 1/\sqrt{2}, \\ 1/\eta, & 1/\sqrt{2} < \eta \leq 1. \end{cases}$$

These are certified construction lower bounds. They do not determine the exact global constant for either topology.

8. AN UNCONDITIONAL $k = 3$ UPPER ENVELOPE

The elementary coefficient two can be improved without a rank-one or scalar reduction assumption. For independent bilinear laws on the binary chain and branching topologies, arbitrary finite dimension and arbitrary orthogonal projector rank, define

$$\eta_* = \sqrt{\frac{\sqrt{5}-1}{2}}, \quad U_3(\eta) = \begin{cases} 1 + \sqrt{1-\eta^2}, & 0 < \eta \leq \eta_*, \\ \frac{\sqrt{1+\eta^2}}{\eta}, & \eta_* < \eta \leq 1. \end{cases}$$

Theorem 8.1 (Tightened projected-root envelope for $k = 3$). *For both declared topologies,*

$$C_{3,\text{ind}}^P(\eta) \leq U_3(\eta) < 2 \quad (0 < \eta \leq 1).$$

In particular $U_3(1) = \sqrt{2}$. The bound follows from the exact local-error telescoping identity and a Pythagorean coupling of the two propagated-error terms at the first internal node.

This replaces the earlier conditional scalar-reduction bookkeeping entry, whose assumptions were not proved. The certified lower curve from the previous section approaches the old coefficient two as $\eta \downarrow 0$, so the new envelope gives a strict finite-defect improvement while retaining

$$\lim_{\eta \downarrow 0} C_{3,\text{ind}}^P(\eta) = 2.$$

9. NON-ATTAINMENT, FINITE SUPPORT, AND THE REMAINING $k = 3$ PROBLEM

Proposition 9.1 (Pointwise non-attainment of the envelope). *No individual configuration in the declared independent-law chain or branching class attains $E^P = U_3(\eta)\rho M^2 L_T$ for $0 < \eta \leq 1$.*

The proof uses a projector-independent self-adjointness argument at the second node and then the incompatibility between the required triangle alignment and the norm-saturation conditions at the root. The endpoint $\eta = 1$ is covered separately because the M9 optimizer is then the interior point $q^* = M/\sqrt{2}$, with both equality contributions nonzero. The corrected proof files are `research/math_closure/k3/m10_non_sharpness_of_m9.tex` and `research/math_closure/k3/m10_endpoint_eta_one.tex`.

Non-attainment alone would not rule out a supremum approached by a sequence. Here the fixed-tree support-compression theorem supplies a uniform finite dimension bound: for each type,

$$\dim W_\tau \leq 2(\text{leaf_count}_\tau + 2\text{node_count}_\tau),$$

which is 20 for a one-type binary $k = 3$ tree (or 22 with proper-projector padding). The resulting finite union of tensor/projector parameter spaces is compact and the objective is continuous. Combining this with M10 gives

$$C_{3,\text{ind}}^P(\eta) < U_3(\eta), \quad 0 < \eta \leq 1.$$

The gap is strict; before the topology-specific M13–M15 refinements below, the support-compression route supplied only a nonquantitative statement. The support-compression proof is in `dimension_rank/fixed_tree_support_compression.tex`.

For the ordered chain, the same cross-term geometry yields a stronger quantitative statement without any saturation assumption. Writing

$$W_3(\eta) = \begin{cases} \sqrt{4 - 3\eta^2}, & 0 < \eta \leq \sqrt{2/3}, \\ \frac{2}{\sqrt{3}\eta}, & \sqrt{2/3} \leq \eta \leq 1, \end{cases}$$

the contraction Gram matrix at the second node gives

$$C_{3,\text{ind},\text{chain}}^P(\eta) \leq W_3(\eta) < U_3(\eta) \quad (0 < \eta \leq 1).$$

The strict inequality is explicit on each interval. This is a chain-only envelope: it does not extend to branching without a separate cross-term argument. The proof is recorded in `research/math_closure/k3/m13_unconditional_chain_envelope.tex`.

The chain envelope is in fact exact. Set $t = \min\{\eta, \sqrt{2/3}\}$, use a two-dimensional rank-one projector, let the first node produce $te_1 + \sqrt{1-t^2}e_0$, and let the second node be the orthogonal two-plane rotation with projected leakage t . A rank-one root map sends the resulting error sum isometrically to the projected root line while having zero root residual. Thus

$$C_{3,\text{ind},\text{chain}}^P(\eta) = W_3(\eta), \quad 0 < \eta \leq 1.$$

The explicit construction and verification are in `m14_exact_chain_constant.tex`. This is complemented by the branching result below.

For branching, scalarize the root law and apply nuclear/operator-norm duality to the two error terms. Their first factors are orthogonal, while the angle of their second factors is fixed by the child residual. This gives the same upper envelope W_3 . A polar-factor root law in dimension two attains the nuclear norm, so

$$C_{3,\text{ind},\text{branch}}^P(\eta) = W_3(\eta), \quad 0 < \eta \leq 1.$$

The proof and executable witness are in `m15_exact_branching_constant.tex`.

10. ASYMPTOTIC SHARPNESS FOR EVERY FINITE BINARY TOPOLOGY

The fixed-eta constants above do not exhaust the binary-tree asymptotic question. For every fixed ordered full-binary tree T with k internal vertices, an explicit two-dimensional real-plane witness (the plane identified with $\mathbb{C}$) closes the asymptotic coefficient. Set $\theta = \arcsin \eta$, use $P(z) = \text{Re}(z)$ and unit leaves, apply $\mu_\theta(z, w) = e^{i\theta}zw$ at each non-root vertex, and use the scalar off-diagonal law

$$\mu_\star(z, w) = (\text{Im } z \text{ Re } w + \text{Re } z \text{ Im } w)1$$

at the root. Complex multiplication makes the ambient value depend only on the number of internal vertices, not on the binary shape, and gives

$$E_T^{\text{proj}}(\eta) = |\sin((k-1)\arcsin \eta)|.$$

The universal projected-root bound therefore yields the theorem

$$\lim_{\eta \downarrow 0} C_{T,\text{ind}}^P(\eta) = k - 1$$

for every fixed finite ordered full-binary topology. This is an asymptotic statement: fixed-eta equality, higher arity, same-law/gated variants, and growing-tree uniformity remain open. The proof and deterministic evaluator are in `m18_binary_tree_asymptotic_sharpness.tex` under `research/math_closure/k3`.

The same phase-counting argument does not depend on binary arity. For a node of arity m , use the real m -linear product law $\mu_{m,\theta}(z_1, \dots, z_m) = e^{i\theta} \prod_j z_j$ at every non-root vertex, and use $\mu_{\star,d}(z_1, \dots, z_d) = \text{Im}(\prod_j z_j)1$ at a root of arity d . Both have norm one; the non-root defect is η and

the root defect is zero. Thus for every fixed finite ordered rooted topology with all arities at least two,

$$E_T^{\text{proj}}(\eta) = |\sin((k-1) \arcsin \eta)|, \quad \lim_{\eta \downarrow 0} C_{T,\text{ind}}^P(\eta) = k-1.$$

The arity-general proof and evaluator are recorded in the M19 dossier.

For $k = 3$, the finite-arity question can be closed at fixed η , not only asymptotically. Freeze all leaf inputs at their unit projected values. A node with one internal child becomes an effective linear law, while a node with two internal children becomes an effective bilinear law; both inherit no larger operator-norm or closure-defect budgets. With three internal vertices the internal-child skeleton is necessarily a chain or a branching root. M13–M15 therefore give the upper bound $W_3(\eta)$. Conversely, inserting unit e_0 gates into the additional leaf slots embeds the dimension-two M14/M15 witnesses. Hence, for every finite ordered arity profile T with $k(T) = 3$,

$$C_{T,\text{ind}}^P(\eta) = W_3(\eta), \quad 0 < \eta \leq 1.$$

This is the M20 fixed-eta all-arity closure; its proof and evaluator are in the M20 dossier.

10.1. Analytic proof of the fixed- η $k = 3$ closure. The fixed- η conclusion above is analytic; the Python evaluator is only a check of the formulas. We record the reduction because it also makes the normalization and the equality mechanism explicit. By homogeneity take $M = L_T = 1$. At the first internal node write the exact decomposition $F_1 = R_1 + D_1$ with $D_1 \perp R_1$, and put

$$q = \|D_1\|, \quad p = \|R_1\|, \quad 0 \leq q \leq \eta, \quad p \leq \sqrt{1 - q^2}.$$

After freezing the sibling leaf at its unit projected value, the next law is a contraction N . For the normalized directions of D_1 and R_1 , let s be the norm of the leaked component of the latter image; then $0 \leq s \leq \eta$. The two propagated contributions at the projected root, the root contraction, and the Gram-matrix estimate

$$|\langle Nu, QNv \rangle| \leq s\sqrt{1 - s^2} \quad (u \perp v)$$

give the exact scalar envelope

$$(20) \quad \frac{E_T^{\text{proj}}}{\rho M^2 L_T} \leq \frac{\sqrt{q^2 + p^2 s^2 + 2qps\sqrt{1 - s^2}}}{\eta}.$$

The Gram estimate follows by applying positivity of $K = N^*QN$ and $I - K$ to the span of u, v : if $K_{vv} = s^2$, then $|K_{uv}|^2 \leq as^2$ and $|K_{uv}|^2 \leq (1 - a)(1 - s^2)$; maximizing over $a \in [0, 1]$ gives $|K_{uv}| \leq s\sqrt{1 - s^2}$. The right side of (20) increases with p , so set $p = \sqrt{1 - q^2}$ and write $q = \sin \alpha$, $s = \sin \beta$. Its square is

$$f(\alpha, \beta) = \sin^2 \alpha + \cos^2 \alpha \sin^2 \beta + 2 \sin \alpha \cos \alpha \sin \beta \cos \beta.$$

With $u = \tan \alpha$ and $v = \tan \beta$,

$$f = \frac{(u+v)^2 + u^2 v^2}{(1+u^2)(1+v^2)}, \quad 4(1+u^2)(1+v^2) - 3((u+v)^2 + u^2 v^2) = (u-v)^2 + (uv-2)^2 \geq 0.$$

Thus $f \leq 4/3$. On the square $0 \leq \alpha, \beta \leq \arcsin \eta$, both partial derivatives are nonnegative up to $\eta = \sqrt{2/3}$, so the maximum is at $q = s = \eta$ and equals $\eta^2(4 - 3\eta^2)$. Above that threshold the square contains the equality point $u = v = \sqrt{2}$, so the maximum is $4/3$. Dividing by η yields

$$W_3(\eta) = \begin{cases} \sqrt{4 - 3\eta^2}, & 0 < \eta \leq \sqrt{2/3}, \\ 2/(\sqrt{3}\eta), & \sqrt{2/3} \leq \eta \leq 1. \end{cases}$$

For the chain, the two-dimensional M14 witness attains both branches. For the branching root, the two first child factors are orthogonal. Scalarizing the root law and using nuclear/operator-norm duality reduces its two-term functional to

$$\sqrt{q^2 a^2 + p^2 b^2 + 2qpb\sqrt{a^2 - b^2}}, \quad 0 \leq b \leq a \leq 1,$$

where a is the norm of the second child value and b its defect. The expression increases with a and p , hence $a = 1$ and $p = \sqrt{1 - q^2}$, recovering the same scalar objective. The M15 polar-factor witness attains it, so both binary $k = 3$ topologies have constant W_3 .

Finally, the M20 finite-arity reduction freezes every additional leaf at its unit projected value. A node with one or two internal children then becomes an effective linear or bilinear law. Multilinearity and projector contractivity preserve the operator-norm budget, while the projected-input closure residual can only decrease. The internal-child skeleton is therefore either the chain or branching skeleton, and the dimension-two witnesses embed by adding these frozen slots. No numerical search is used in this reduction.

The same-law question has a precise scope distinction. If a single bilinear law is repeated but the projected leaf vectors and projector rank are allowed to vary, orthogonal tags can select blocks of one direct-sum law. M21 uses this construction to simulate the M14 chain witness and the M15 branching witness without increasing the operator-norm or closure-defect budgets. Thus for the tagged same-law binary class,

$$C_{3,\text{tag-same}}^P(\eta) = W_3(\eta), \quad 0 < \eta \leq 1.$$

This does not assert the same value for rank-one/common-leaf same-law or gated-planar subclasses.

There is nevertheless an exact regime inside the strict rank-one/common-leaf class. M22 repeats a single rotation with leakage $\alpha = \sqrt{2/3}$, common leaf e_0 , and $P = \text{span}(e_0)$. For every $\eta \geq \alpha$ the witness is admissible and its normalized error is $2/(\sqrt{3}\eta) = W_3(\eta)$. The M14 upper bound therefore proves exact equality for this high- η chain regime. The low- η regime and the corresponding branching class remain open.

The remaining strict chain class admits an exact reduction even though its low- η value is not yet evaluated. For one repeated bilinear law μ , common projected leaf e_0 , and $P = \text{span}(e_0)$, define $Ax = \mu(x, e_0)$. The ambient chain is $A^3 e_0$, whereas the recursively projected chain is $PAPAPe_0$. Since $PAP = \langle e_0, Ae_0 \rangle P$,

$$E^P = |\langle e_0, A^3 e_0 \rangle - \langle e_0, Ae_0 \rangle^3|.$$

Conversely, every finite-dimensional contraction A is realized by the gated repeated law $\mu_A(x, y) = Ax\langle e_0, y \rangle$. Thus M23 gives an equivalent contraction-operator supremum for the strict class. It is a reduction theorem, not a claim that the low- η supremum has been solved; branching and broader same-law/gated classes remain outside this statement. The exact proof and evaluator are recorded in `research/math_closure/k3/m23_rank_one_same_law_chain_operator_reduction.tex`.

11. NODEWISE AND TOPOLOGY-AWARE DIRECTIONS

The exact subset expansion supports mixed-mask dynamic programming, source path sums, and scalar telescoping-order optimization. These certificates attribute responsibility but do not imply a universal topology-only sharp constant. Chain, branching, depth, fan-out, fan-in, and operator-sharing patterns must be treated as separate coordinates of the extremal problem. The fixed finite arity-compatible asymptotic statement is now proved by the preceding construction, and M20 closes its fixed- η specialization at $k = 3$. Fixed- η independent-law sharpness for $k \geq 4$ and growing-tree uniformity remain open rather than being inferred from finite-tree results.

TABLE 1. V5 sharpness status.

class	result	status
$k = 2$, independent laws	$C_{2,\text{ind}}^P(\eta) = 1$	exact under assumptions
$k = 2$, explicit repeated map	$C_{2,\text{same-law}}^P(\eta) = 1$	exact under declared class
$k = 2$, canonical gated rotation	$E^P = \eta^2$	exact narrower value
$k = 2$, contractive gated repeated law	$C_{2,\text{gate-contr}}^P = 1$	exact under explicit definition
$k = 3$, independent chain	stated analytic lower bound	certified lower bound
$k = 3$, independent branching	stated analytic lower bound	certified lower bound
$k = 3$, independent chain/branching	$L_3 \leq C_{3,\text{ind}}^P = W_3 < U_3$	exact independent-law values
$k = 3$, independent chain	$C_{3,\text{ind,chain}}^P \leq W_3 < U_3$	explicit quantitative gap
$k = 3$, independent chain	$C_{3,\text{ind,chain}}^P = W_3$	exact fixed- η value
$k = 3$, independent branching	$C_{3,\text{ind,branch}}^P = W_3$	exact fixed- η value
finite arity-compatible tree, independent laws	$\lim_{\eta \downarrow 0} C_{T,\text{ind}}^P(\eta) = k(T) - 1$	exact asymptotic value
$k = 3$, all finite arity profiles, independent laws	$C_{T,\text{ind}}^P(\eta) = W_3(\eta)$	exact fixed- η value
$k = 3$, tagged same-law binary	$C_{3,\text{tag-same}}^P(\eta) = W_3(\eta)$	exact fixed- η value
$k = 3$, rank-one same-law chain, $\eta \geq \sqrt{2/3}$	$C = W_3(\eta)$	exact high- η regime
$k = 3$, strict rank-one same-law chain	exact unary-operator reduction	value open

12. LIMITATIONS AND CONCLUSION

The results are finite-dimensional and scoped. The exact value for the explicit repeated-map class does not close every repeated-law subclass; variable-gate/arbitrary-leaf gated variants, low- η rank-one/common-leaf same-law, rank-one/common-leaf branching, fixed- η independent-law sharpness for $k \geq 4$, uniformity over unbounded tree growth, globally tight multilinear norms, theorem-level novelty, and independent human review remain open. The paper therefore reports a classification at $k = 2$, exact independent-law constants for every finite $k = 3$ arity profile, certified constructions, and a precise map of the remaining extremal frontier.

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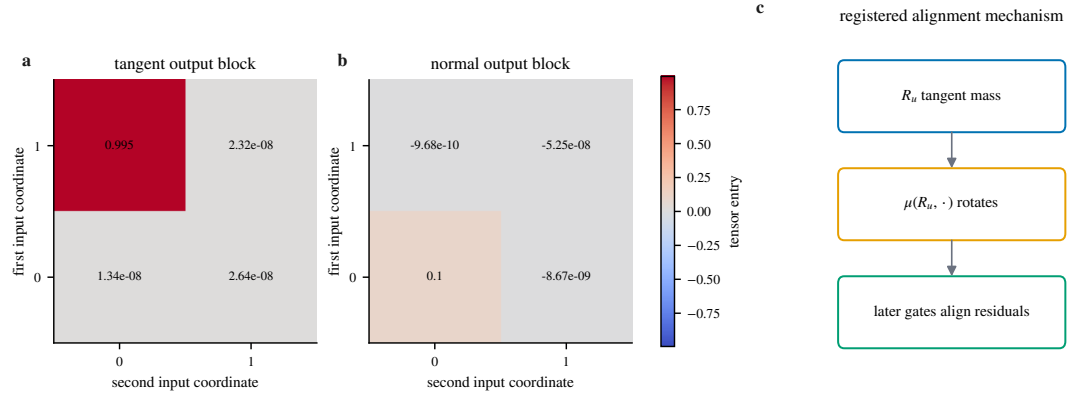


FIGURE 1. Registered projected-tree extremizer geometry reused from the finite v4 evidence package.